GREENPOINT
Technologies

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G1632-SRD-4403

SUPPLIER REQUIREMENTS DOCUMENT–

IN-FLIGHT ENTERTAINMENT/CABIN MANAGEMENT SYSTEM/GSM

Revision A

B787-9 Interior Modification

Greenpoint Technologies, Inc.

CAGE Code 1V8T8
Bothell, WA 98011

Revision Status

Rev	Description	Approval
IR	INITIAL RELEASE	ENGR: NELS LIDDIATT PE: JERRY ANDERSON 5/19/2022
A	Ref: PU 021753 Changes are indicated in blue test throughout and are summarized below. Updated Acronyms and Abbreviations 1.3.4 Updated Preliminary ATP document requirements 1.3.7 Added Both LHT and PAC AMM requirement 2.1.1 Added light default 2.2.3 Updated Boarding Music function 2.2.5 Added clarity for room tablet's function 2.2.6 Updated Customer Logo requirement 2.2.10 Added section 2.3.2 Added AeroSuite PCU requirements 2.4 Added note for clarity for Seat Actuation 2.6 Added Accent down lighting function 2.10 Removed Business and Economy class EDW controls 2.12 Updated new components for 1632 program Appendix C- Added Monitor controls Appendix D & F- Update monitor sizes Appendix E- Update Lighting matrix's for all rooms	ELECTRONIC SIGNATURES ON FILES. SEE FRONT PAGE FOR APPROVER AND DATE.

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List of Acronyms and Abbreviations

Acronym	Definition
Aux	Auxiliary
A/V	Audio / Video
AVOD	Audio/Video On Demand
BGM	Boarding Music
BITE	Built In Test Equipment
BRT	Bright
CFR	Code of Federal Regulations
CMM	Component Maintenance Manual
CMS	Cabin Management System
CFR	Code of Federal Regulations
CT	Crew Terminal
DER	Designated Engineering Representative
DFCU	Dual Frequency Converter Unit
DND	Do Not Disturbed
EASA	European Aviation Safety Agency
ECBU	Electronic Circuit Breaker Unit
ECS	Environmental System
FAA	Federal Aviation Administration
FAR	Federal Aviation Regulations
FCR	Final Concept Review (milestone)
FWD	Forward
GRD	Graphic Requirement Document
GTI	Greenpoint Technologies, Incorporated
GUI	Graphical User Interface
GSM	Global System for Mobile Communications
HAS	Hardware Aspects of Certification
HSV	Hue;Saturation;Value
HVP	Hardware Verification Plan
IFE	In Flight Entertainment
IR	Initial Release
IWG	International Water Guard
JDP	Joint Development Phase
LCD	Liquid Crystal Display
LED	Light-emitting Diode
LRU	Line Replacement Unit
MS	Modification Specification
MSP	Membrane Switch Panel
OEM	Original Equipment Manufacturer
PA	Public Address
PAX	Passenger
PMA	Parts Manufacturer Approval
PCU	Passenger Control Unit
PED	Passenger Electronic Devise

Acronym	Definition
PHAC	Plan for Hardware Aspects of Certification
PRAM	Precorded Announcement
RFQ	Request for Quote
RGB	Red, Green, Blue
RTCA	Radio Technical Commission for Aeronautics
SRD	Supplier Requirements Document
Spec	Specification
STC	Supplemental Type Certificate
TSO	Technical Standard Orders
TBD	To Be Determined
TTOL	Taxi, Take Off, and Landing
VIP	Very Important Person
WAP	Wireless Access Point

1 General Description

1.1 Description

Ref Mod Spec G1632-MS-01, Deck Plan G16320789

This document provides requirements for the In-Flight Entertainment/ Cabin Management System/eXPhone to be installed and certified concurrent with GTI's VIP interior configuration on (2) Boeing *787-9* aircraft.

The delievered aircrafts are configured with a commercial interior configuration for Suparna Airlines as shown in Figure 1 with an existing eX3 IFE and GCS (w/ eXConnect) Panasonic systems.

The VIP modification will reconfigure the existing IFE components along with new IFE/CMS/GCS components.

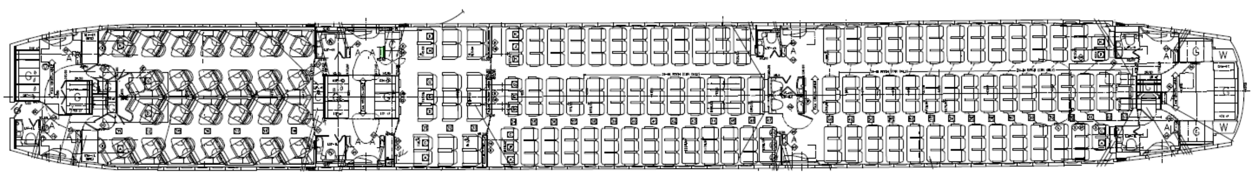


Figure 1 – Induction Configuration

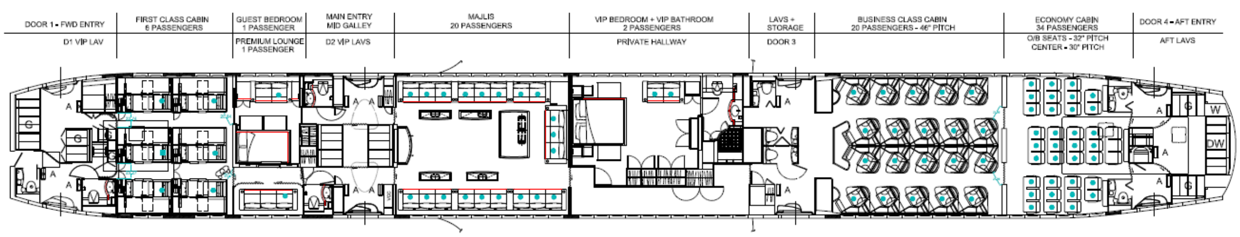


Figure 2 – Proposed VIP Deckplan

The In-Flight Entertainment/ Cabin Management/ GSM System has several purposes, the most notable of which are:

1) The system shall have a Digital Audio and Video distribution system and accept and distribute multiple A/V sources throughout the passenger cabin as defined within this document.
2) The system shall provide the passenger and crew with control over the cabin environment.
3) The system shall alert the flight Attendant of abnormal conditions within the cabin.
4) The system shall have the capability of interfacing with 3rd party supplied systems for command, control and status monitoring for systems such as: Entertainment systems, Lighting systems, Pleated Shades and other systems as defined in this document.

5) *The system shall have user friendly Built-In Test Equipment (BITE) to troubleshoot and diagnose system failures.*
6) *The system shall provide a GSM system for communications on/off wing.*

Note: This version of the document contains high level requirements to show expected In-Flight Entertainment/ Cabin Management System interfaces/ Global System for Mobile Communications. Not all functional requirements internal to the systems or detail design requirements are necessarily contained within this document. In-Flight Entertainment/ Cabin Management System/ Global System for Mobile Communications will be expected to be able to support functions internal to the system within the capabilities of the equipment. The detail system requirements and further system definition will be further matured during the Joint Design Phase.

1.2 Work Scope

1. Effort shall include design, development, qualification, certification, and manufacturing services.
2. Supplier will be responsible for providing electrical/electronic components, software, data drawings and all certification data required to support the In-Flight Entertainment/ Cabin Management System design and integration effort.
3. Full testing of the systems with exact representative components and quantities shall be accomplished in a laboratory or other off aircraft environment prior to delivery of system software. Deviation will need to be approved by GTI.
4. IDAIR will be responsible for interconnect wiring diagram between In-Flight Entertainment, Cabin Management and GSM system components.
5. IDAIR will be responsible for Acceptance Test Plan document for complete system test.
6. IDIAR will be responsible for Aircraft Maintenance Manual and Fault Isolation Manual supplements.
7. Certification & Approvals:
    a. Quality:
        i. Suppliers shall be compliant to AS9100:2016 standards.
    b. Parts:
        i. Parts must be accompanied with airworthiness tag (i.e., PMA, TSO, EASA Form 1, and/or 8130-3). Parts without airworthiness tag will be subject to onsite conformity at vendor's facility.
    c. Engineering & Certification Data:
        i. Vendor shall furnish all data required to support GTI Engineering design, including certification of products where applicable and STC process. In the event data cannot be released as to safeguard proprietary information, vendor to authorize GTI or a delegate by GTI to review the data at vendor's facility.
    d. Flammability:
        i. Flammability substantiation per FAR 25.853 (a) by means of similarity, analysis, or testing, shall be responsibility of vendor.
    e. Safety / Reliability

i. Vendor shall provide analysis or service data to support reliability of product or specific functions in order to support product certification
ii. If product has safety critical features the vendor shall provide supporting documentation to justify design rationales, design assurance, and reliability as required. This may include Safety Assessments, Failure modes and effects analysis, Fault Trees, Hazard Assessments, and reliability assessments.

1.3 Project Milestones

1. Kickoff/Initial Technical Coordination Meeting
    a. Preliminary System Block/Interface Diagram
    b. Part List
    c. Recommended Spare Parts list (RSPL)
    d. Preliminary port assignment and interface description documents
    e. Preliminary equipment outline drawings, which include weights, envelops, Mechanical & electrical interface, installation requirements & limitations, LRU electrical loads
2. Preliminary Design Review
    a. Preliminary port assignment and interface description Doc
    b. Buyer Furnished Equipment (BFE) List
    c. PMA supplement (for all components with PMA)
    d. Preliminary Customer Requirement Document (1 week after PDR)
    e. Preliminary System Definition Spreadsheet
    f. Preliminary GRD design
3. Component Critical Design Review
    a. Released System Block/Interface Diagram
    b. Released equipment drawings
    c. Flammability Test Plan
    d. Qualification Test Plan
4. Software Critical Design Review
    a. Released System Definition Spreadsheet
    b. Released Customer Definition Document (1 week after CDR)
    c. Preliminary Acceptance Test Plan Document (3 months before Customer ATP)
    d. GRD design
5. Final testing & Qualification
    a. Drawing List (Excel Overview of summitted Documentation)
    b. Flammability Test Report
    c. Qualification Test Report
    d. PMA Supplement (for all components pending PMA)
6. Acceptance Test Plan (Customer Test)
    a. Released Acceptable Test Plan Document (4 weeks before Customer test)
    b. Approved and Accepted Customer Test
7. Customer Acceptance & Delivery of Tech Pubs
    a. Aircraft Maintenance manual (AMM) Supplement (LHT and PAC)
    b. Fault Isolation Manual (FIM) Supplement

c. Training manuals

1.4 Aircraft Environment

All of the IFE/CMS/GSM equipment will be installed in the pressurized areas. Equipment will be located in multiple locations throughtout the aircraft. These locations include; Cabin Monuments with limited cooling, Overhead areas and below floorboards.

2 Design Requirements

2.1 General System Requirements

The IFE system shall have a Digital Audio and Video distribution system and accept and distribute multiple A/V sources throughout the passenger cabin as defined within this document.

The CMS/IFE system shall have the capability of interfacing with 3rd party supplied systems for command, control and status monitoring for systems such as: Entertainment systems, Lighting systems, Connectivity and other systems as defined in this document.

2.1.1 System Power: Start-Up and Reset

The IFE/CMS system shall be able to withstand/sustain up to 200ms momentary power interruption without disruption or noticeable effect by the passenger. This means at a minimum that any critical components must not have to go through a boot up sequence, Ethernet switches or WAPS shall not reset and all settings must be immediately returned to the previous state after power is restored.

All CMS controlled lighting shall be defaulted to Boarding light scenarios upon initial start-up. Lighting shall remain at its current state in case of a power interruption.

2.1.2 Lip-Sync

As per the ATSC Implementation Subcommittee IS-191: the audio associated with a Video program shall not lead the video by more than 15ms nor lag the video by more than 45ms.

2.1.3 Monitor Resolution

All monitors and displays shall be 1080P or higher resolution.

2.1.4 Wi-Fi Streaming

The system shall have multiple SSID's networks that provide dedicated bandwidth requirements for the priority SSID network. The system shall provide enough bandwidth to allow 84 passengers and 12 Crew to stream to PED and room tablet controls. The bandwidth performance detail requirements will be defined during the Joint Design Phase.

2.1.5 Aux HDMI Port

The system shall provide HDMI ports that support carry on equipment (Apple TV, Roku, AirMont TV, etc.) that can be displayed locally and/or at multiple location throughout the aircraft.

2.2 Functional Requirements

The CMS/IFE System shall provide all the modules necessary to interface to the applicable systems and sub-systems in order to perform the functions described in the body of this document and appendices.

2.2.1 Audio/Video Equipment and Distribution

The system will distribute a variety of audio and video sources. The audio source shall be synced with the video selection unless specified below. However, the user shall be able to change the audio selection to another source independently if desired.

See Appendix D for the list of A/V equipment and distribution.

2.2.2 Default States

Audio/Video default states. Other system default states listed in appropriate sections or appendices.

Default states upon IFE/CMS system power on		
Monitor power	See A/V matrix	-All Attendant Crew Terminals will default on and will display the control GUI menu -If applicable, when default state is “On” any monitor lifts will be activated and translating panels opened to present monitors. Otherwise all lifts remain in down position and translating panels will be closed. If the lift or translating panel is manually operated the monitor will remain off until deployed or opened.
Video selection		Bulkhead monitors will default to logo background screen, seat back monitors will default to GUI home page.
Audio selection		Muted
Notes:		

2.2.3 Boarding Music & Zonal Music

The system shall be capable of playing Boarding music over all PA speakers with a simple command initiated from the Attendant Crew Terminal (CT). The Boarding Music command will initiate Boarding Music over the PA speakers.

2.2.4 CMS Status Inputs/Outputs

The status inputs are from aircraft systems or third-party equipment to provide information to the CMS/IFE system which can be displayed on the CT or used to initiate another action. These inputs will be defined as the system is fully defined.

2.2.5 Wireless device support

The system shall utilize wireless PED devices as Crew, Room and passenger controllers. The equipment and operating systems must be supported are listed below. Preferred as an application to each device.

Each room tablet shall have a separate URL address for their assigned room. Tablet startup shall default to the last assigned room URL.

Hardware	Screen Resolution/App	Operating System
Apple Devices (iPad Air, 2022 version)	2360x1640	iOS

2.2.6 Moving Map System –

The system shall distribute a Moving Map system as specified below.

Vendor:	IDAIR
Model:	Betria
Default audio when selected:	☒None/Mute ☐Previously selected (do not change audio) ☐Other _______________
☒Interactive mode controlled by IFE	Shall be able interactive on seat monitors/PED tablets.
☐Custom logo	Same as PFE
☒Multiple language support	English, Arabic
☒City names shown in multiple languages _______________	English, Arabic
☒Mecca locator	Relative Location Indicator showing Mecca direction at a minimum
Units to be displayed in: ☒Standard ☒Metric	Units cycled after each script
☒Relative location indicator	
Aircraft livery	Paint scheme will be provided

2.2.7 PA System

The system shall integrate with the retained OEM system as specified below.

	CMS receives keyline from PA - Active	Keyline removed from CMS PA – In-active
Speaker audio	Mute	Volume returns to previous level
Headphone audio (if applicable)	-PA audio broadcast over headphones - During a PA announcement the Passenger Control Unit will indicate PA is active and lock controls	Volume returns to previous level
All controlled audio sources. NOTE: not all sources are controlled (e.g. AUX Source inputs) these will not be affected.	Pause	Programming returns to previous state
All controlled video sources. NOTE: not all sources are controlled (e.g. AUX Source inputs) these will not be affected.	Pause	Programming returns to previous state
A/V Selections	Unavailable, locked	Selections available
AVOD server shall be capable of providing boarding music functionality	Pause	Programming returns to previous state

2.2.8 Call system

Calls from the Cockpit and OEM Lavatories will not be indicated on the Attendant CT's. All other calls must be indicated on the Attendant CT's and Crew tablets including calls from all VIP Lavatories, MSP's, PED Tablets, Passenger control units and Room Tablets.

Passenger Call Activation/De-activation – A passenger call is activated by pushing one of the Call buttons on one of the call devises located throughout the cabin. The call can be de-activated by pressing the same button a second time. The status on the call device will highlight indicating the button was pressed and will stay illuminated until the call is de-activated either by the attendant or local reset. When a call is pressed a chime will be actuated over PA speakers defined by GTI.

Indication – Upon a PAX Call being activated at any given location, a PAX icon call will pop up on the CT's and Crew tablet Main Menu GUI screen. PAX Call will be displayed in multiple colors by priorities with passenger description only, no LOPA identifiers. Any subsequent PAX calls will be displayed by priority. Once a PAX icon is pressed on the Call Request menu, the GUI will automatically display the

PAX Call location. If the PAX call is cleared, then all remaining PAX call icons will be re-arranged in order of priority.

VIP areas will have priority call requests over other zones. These priority calls will be displayed above all other calls.

2.2.9 Decompression System

- The CMS/IFE system must be capable of receiving a Decompression event signal from the aircraft
- When a decompression signal is received all Ceiling lights will turn on to the bright setting
- Monitors will turn off or backlight turn off.
- 110VAC outlets will be commanded off except for the outlets that are configured to Flight Deck Mode.

2.2.10 Economy Bulkhead Monitors

- The Crew Terminal/Crew mode Tablets shall control the (3) three Economy Bulkhead Monitors On/Off via a soft toggle switch.

2.3 Graphical User Interface (GUI)

2.3.1 Language Support

Languages	
List languages supported	English, Arabic
Default language	English

The GUI shall provide language selection between English and Arabic on the following devices/systems:

- Seat Back Monitors
- Passenger Control Units
- Tablet in IFE Mode
- Maps

All text from all languages listed above must fit within the confines of buttons or other graphic features.

2.3.2 General Settings

The following cabin control devices shall automatically turn off or Dim the backlight after a configurable duration, as specified below:

Devise Type	Backlight Dim Timeout	Backlight Timeout
Passenger Control Units	N/A	30 Seconds
AeroSuite Passenger Control Units	N/A	1% Display Brightness after 30 Seconds
Room Control Panels	N/A	30 Seconds
Crew Terminals	N/A	2 Minutes
Seat Monitors	N/A	2 Minutes (Unless Video/Audio is displayed)

Tapping the touch screen will reactivate the devise but no key press action will be activated. The only action is to wake up the screen and be ready to process the next command key press.

2.3.3 Customized Controls Per Seat/Screen

Room control panels home screen shall not require no more than 2 inputs to operate simple functions (Shades, Volume, Passenger Call, etc.). Home screen shall have a room input that navigates to the detailed control menu for advance controls. Controls shall be specific for each seat and the GUI shall be customized to remove features not available for certain seats/room.

Passenger control panels shall not require no more than 2 inputs to operate simple function (Shades, Lights, Volume, Passenger Call, etc.). Controls shall be specific for each seat and the GUI shall be customized to remove features not available for certain seats.

2.3.4 Tablet Applications

The following Tablet applications shall be Crew Mode, Room Mode and Passenger Mode (limited seat mode only, See Appendix G). GUI's shall be set to the defined Control Matrix (Appendix C).

2.3.5 Lighting Control

The GUI shall provide lighting control by area with pre-defined scenes and/or per individual light group type. Step less dimming shall be provided for dimmable lights.

Colored (RGB) lighting shall be set to the defined color and intensity per the Lighting Matrix (Appendix E) for each scene. Colored lights shall also be controllable dynamically without setting or changing a scene with the use of a color changer menu.

Advanced colored lighting shall include:

- Adjusting lighting colors via sliders dynamically to set a color
- Saving color settings – light colors may be selected, and the setting can be saved to a color settings list. Within the pop-up pressing the Save button will display a text box to enter in the custom color name. After saving the setting is saved to the settings list making it accessible from other rooms.
- Delete – color settings may be deleted

2.4 Interfacing Systems Requirements

Table 1: Interfacing Systems

ATA	System	Requirement	Comments	Equipment Supplier (TBC)
23	Public Address	Shall interface with PA audio to support boarding Music and key line to mute entertainment audio	CSS interface	Boeing
24	Power Distribution control	Shall provide control and interface for the ECBU's and DFCU (See G1632-SRD-03 for details)	RS-485	Astronics
25	Seat Actuation	Shall provide seat power and interface for Electrical Control Unit Note: Seat power only for Aerosuite	RS-485/115VAC	AeroSuite/ Astronics BC/Collins EY/RECARO
26	Smoke Detection	Shall provide a means to show status indication of each smoke detector when activated. Shall provide a soft button to silence the cabin horn.	VIP Lavatory and VIP areas. Discrete	Kidde
33	Lighting	Shall provide a means of controlling lights.	RS-485/Discrete	BE Aerospace
33	AeroSuite Do Not Disturbed (DND) Light	Shall provide a means to control the DND light in each AeroSuite	Discrete	Collins/ Emteq
33	Carry On Equipment Lights	Shall provide a means to controlling the equipment. On/off control and status.	Discrete	WASP
42	OEM Standard Signals/CSS	Shall provide a means to interface with standard aircraft status signals and use information as needed throughout the system design and certification.	Aircraft signals (RPDU, EDW, WOW, Cruise Phase, Passenger Request, DECOMP, ARINC 429, etc.)	Boeing
52	Pocket Door	Shall provide a means to show status indication of each door	Discrete signal from TTOL critical doors	GTI
56	Window Shades	Shall provide a means of controlling the shades Up/Down by aircraft/Room/Individually.	Discrete	ATG

2.5 Smoke Detector Annunciation

The passenger cabin will be equipped with smoke detectors. When a smoke detector is activated the location will automatically be displayed on the Attendant CTs and Crew mode tablets. The LOPA with smoke detector map will automatically be brought up so the attendant will know a smoke detector is active. A horn silence soft button on the Attendant CT's and Crew mod tablets will be able to silence the cabin horn. A new smoke event will retrigger the horn and smoke detector indication.

2.6 Cabin Lighting

The system will contain LED lighting in the VIP areas. The system will control the LED lighting individually or by lighting scheme by the CMS at local switch panels, touch screens, tablets and at the Attendant CTs.

LED lighting in VIP areas shall feature the following lighting types (see table below):

- Ceiling Accent RGB
- Ceiling Dome RGB
- Sidewall Wash RGB
- Sidewall Accent RGB
- Toekick/Night RGB
- Accent Dome White
- Accent Down RGB
- Reading White
- Mirror White
- Sconce White

Room	AeroSuite	Premium Lounge	Guest bedroom	Majlis	Master Bedroom	VIP LAV	Hallway
Dome Lights			X	X	X	X	X
Reading Lights/ Accent Dome	X		X	X	X	X	
Sidewall Wash Lights			X	X	X	X	X
Ceiling Accent			X	X	X	X	
Accent Down	X						
Toe-Kick Lights/ Night Lights	X	X	X	X	X	X	
Mirror Accent light						X	
Monitor Accent Lights				X			
Shower Dome Light						X	
Carry-On Lamps		X	X		X	X	

2.7 Carry On Equipment (Lights)

The system will contain carry on Sconce and Mirror lighting in the defined VIP area. The system will control the lighting individually or by lighting scheme by the CMS at local switch panels, touch screens, tablets and at the Attendant CTs.

2.8 OEM Standard Signals

The system shall interface with the OEM aircraft systems to receive all the required signals to meet the specification functions and certifications within this document.

2.9 Pocket Door Indication

The system will contain a switch at each pocket to provide pocket status. The IFE/CMS system shall be capable of indication the position of each door. Indication during TTOL phases of flight will indicated red if a door is not in the correct position and green if door is in the correct position on the crew terminal and crew mode tablet. Flight deck indication for a door not in TTOL position, will be active for each phase of flight with the exception of Phase 7 (Cruise).

2.10 Window Shades

The system will contain both OEM Electrical Dimming Windows (EDW) through the whole cabin zones and electrical pleated window shades in the AeroSuites, Premium Lounge, Guest bedroom, VIP Bedroom and VIP lavatory zones.

2.10.1 Electrical Dimming Windows (EDW)

The OEM shades will be retained in their original location and CSS database shall be updated to re-group the EDW's to match the new LOPA layout. New EDW regrouping is shown in Appendix H. EDW controls shall be controlled from all control panels located within the EDW grouping area.

2.10.2 Pleated Window Shades

The pleated shades are made up of two components, a sheer/translucent shade and a blackout shade connected by a bar. The two components operate consecutively so that to close the blackout shade the user must first close the sheer shade.

The Pleated shades will be able to be controlled with one control button to either Fully closed or Fully open. If the control button is pressed again the shades will stop at their current position.

2.10.3 IFE/CMS Shade controls

The IFE/CMS system shall control both the EDW's and pleated windows in all the zones separately or as a group. The shades shall be capable of being controlled as follows:

- By area
- By Left / Right within an area
- Full Cabin
- By Left / Right full cabin

The GUI on the Attendant CT's and tablets (crew mode) shall provide controls to select and control the EDW's and pleated window shades by zone, left or right within each zone and All zones. Zone control for the EDW's and pleated shades shall be made from the room control panel and tablets (room mode).

Note: Business class and economy class zones shall not include EDW controls from the crew terminal.

The GUI on the passenger control panels shall provide controls of the EDW and pleated window shades local to each seating zone. Example: Both Seat 21K and 22K passenger control panel shall control the group 21K EDW's and Pleated window shades. Seat 18G shall include shade control for group 16K shades.

EDW's shall be controlled by 5 different selection modes (0%, 25%, 50%, 75% or 100%).

Pleated shades shall be controlled by two modes referred to as Tap On/Tap Off (TOTO), the other as the "Sustained" mode. These modes are recognized by the CSBS according to the duration of switch closure. Depression of a switch for < 0.5 seconds is interpreted as TOTO, which results in continuous shade movement until shades are completely open or closed. During a TOTO operation the switch can be depressed again to stop shade movement.

Switch depression for a period ≥ 0.5 seconds is recognized as the Sustained mode. In this mode shade travel continues as long as the switch is held depressed, or until shade movement reaches the next intermediate stop. For example, starting with both shades in the open position, depressing and holding the switch causes the translucent shade to move toward the close position. It continues moving toward that position as long as the switch is depressed. When the close position is reached no further shade movement occurs, even if the switch is still depressed. Release of the switch prior to reaching the close position stops shade movement at that point.

2.11 Power Distribution Control and Display

The IFE/CMS system shall be capable of indication, status and control of the Astronics Electronic Circuit Breakers Units (ECBU) and Dual Frequency Converter Units (DFCU) as defined in G1632-SRD-03 document.

2.12 Components / Equipment List

Following are the new components that have not been used on previous GTI 787 programs.

Table 2: Components / Equipment List

Qty	Part Number	Description	Supplier
12	1303-377	Outlet	Panasonic
6	APA3012-001-001	Audio Power Amplifier	LHT
6	LCD3213-001-001	32” LCD Monitor	LHT
6	RD-FA6015-01	USB-C Outlet	Panasonic
1	RD-AA8190-01	Cell Modem	Panasonic
1	RD-AA903687-01	Bracket Assy W/Antenna	Panasonic
1	RD-NB4321-01	BTS	Panasonic
1	4KS420-001	42” OLED Monitor	Rosen
2	4KS650-001	65” OLED Monitor	Rosen

2.13 Obsolescence

a. Equipment selected shall have no obsolescence within the next 10 years.

2.14 Appearance and Finish

b. Bezels and trims shall be plateable to match cabin interior finishes.
c. GUI design shall be customized to match cabin interior design.

2.15 Tech and Maintenance Access

a. A means must be provided to access maintenance features of the system by either maintenance ports or wireless access to the system to obtain maintenance data or software updates.

2.16 Thermal and Acoustic Install

a. All equipment shall be passively or internally fan cooled. Forced air cooling equipment will require to be evaluated. No liquid cooled equipment shall be permitted.
b. Equipment with Lithium Ion Batteries will require to be evaluated. Equipment with lithium batteries requires disclosure such that the certification and installation aspects can be addressed.
c. All equipment shall minimize acoustic noise emissions. Maximum emission is 50dbA for cabin installed equipment.

2.17 Decompression Venting Requirements

N/A

2.18 Specific Requirements

N/A

3 Installation Method

3.1 Accessibility / Clearance

a) Must be able to be installed/removed using standard tools.
b) All Line Replaceable Units (LRUs) must support a removal time of 30 minutes.
c) All equipment installation envelope can be meet.

3.2 Space Reservation / Interfaces

a. Component specifications, TBD during Joint Development phase.

4 Engineering Verification Services

4.1 Stress Evaluation

a) Equipment with moving parts shall be evaluated.

4.2 Weight Information

Table 3: Weight Information

TBD during Joint Development phase

Original Estimate	
Current	
Performance to Target	

4.3 Component Design Standards

4.3.1 Equipment/System Design

a. Functions and design features associated with IFE/CMS are expected to be DAL E.
   i. Functional reliability must be demonstrated as appropriate for DAL
b. The following standards apply to the design:
c. Flammability:
   i. Flammability substantiation per FAR 25.853 (a) ix F, by means of similarity, analysis, or testing, shall be responsibility of vendor.
d. Safety / Reliability
   i. Vendor shall provide analysis or service data to support reliability of product or specific functions in order to support product certification
   ii. If product has safety critical features the vendor shall provide supporting documentation to justify design rationales, design assurance, and reliability as required. This may include Safety Assessments, Failure modes and effects analysis, Fault Trees, Hazard Assessments, and reliability assessments.
e. Structural Substantiation:
   i. Addressed within DO-160G for electrical components.
f. Configuration Control
   i. All drawings, data, hardware design, software design, system design and the associated interoperability and/or system configurations shall be configuration controlled.
   ii. Master System Configuration DL or equivalent for all development/test systems configurations is required. Format to be developed and agreed upon during JDP.

4.3.2 Equipment Qualification (DO-160G)

a. Electrical components shall be compliant with each condition of DO-160G (Reference Qualification Requirements in appendix A) by means of similarity, analysis, and/or test.
b. Installation location is Aircraft cabin pressurized
c. DO-160 testing, if required, shall be the responsibility of Supplier
d. Supplier shall be responsible for submitting DO-160G substantiation data in support of GTI STC.
e. DER approval of substantiation data shall be the responsibility of Supplier.

4.3.3 Software Design (DO-178C)

a. System software shall be compliant with RTCA DO-178C by means of similarity, analysis, and/or test, unless Supplier can justify function controlled by software do not affect safety along with Level E software.
    a. Level E software shall still be configuration controlled
b. DO-178 testing, if required, shall be the responsibility of Supplier
c. Supplier shall be responsible for submitting substantiation documentation (i.e., Plan for Software Aspects of Certification (PSAC), Software Acceptance Test Plan/Report, System Safety Analysis, etc.) to support GTI STC.
d. DER approval of substantiation data shall be the responsibility of Supplier.

4.3.4 Electronic Hardware Design (DO-254A)

a. Complex hardware shall be compliant with RTCA DO-254, latest revision, by means of similarity, analysis, and/or test, unless supplier can justify complex hardware do not affect safety.
b. DO-254 testing, if required, shall be the responsibility of Supplier
c. Supplier shall be responsible for submitting DO-254 substantiation documentation (i.e., Plan for Hardware Aspects of Certification (PHAC), Hardware Verification Plan (HVP), and/or Hardware Accomplishment Summary (HAS), etc.) to support GTI STC.
d. DER approval of substantiation documents shall be the responsibility of Supplier.

5 Documentation

5.1 Applicable Documents

The following documents provide sources for definition of the content herein.

Table 4: Applicable Documents

Applicable	Number	Document Description	Rev or Date
Yes	G1632789	Deck Plan	Rev L
Yes	G1632ECMUS10	Engineering Component Mark Up Sheet	Rev IR
Yes	G1632-SRD-03	Supplier Requirements- Power Distribution Control and Display Unit	Rev IR

5.2 Related Documents

The following industry documents provide derived requirements as applicable.

Table 5: Related Documents

Applicable	Number	Document Description	Rev or Date
Airworthiness Requirements			
Yes		FAR Part 25 [Amdt 25-123]	See Cert Plan
Boeing Specifications			
YES	D6-36440	Standard Cabin System Requirement Document	E or newer
Industry Specifications			
YES	RTCA DO-160G	Environmental Conditions and Test Procedures for Airborne Equipment	G
YES	RTCA DO-178C	Software Considerations in Airborne Systems and Equipment Certification	C
YES	RTCA DO-254A	Design Assurance Guidance for Airborne Electronic Hardware	A

5.3 Data Requirements

See Appendix B - Vendor Data Requirements checklist

Appendix A - Qualification Categories

This is the proposed test categories for the GTI qualification baseline, any deviations to this baseline vendors should make comment in the vendor comments column and those deviations will be considered on a case by case basis.

Test Description	DO-160 Test Section	Proposed Test Category (DC Eq.)	Proposed Test Category (AC Equip.)	Vendor Comments*
Ground Survival Low Temperature & Short-Time Operating Low Temperature	Section 4.5.1	A1	A1	
Operating Low Temperature	Section 4.5.2	A1	A1	
Operating High Temperature	Section 4.5.3	A1	A1	
Altitude	Section 4.6.1, (15,000')	A1	A1	
Temp Variation	Section 5.3.1	C	C	
Humidity	Section 6.3.2	A	A	
Operational Shocks Test Procedure or Alternate Test Procedure	Section 7.2.1 or 7.2.2	A	A	
Crash Safety Impulse (or) Alternate Test Procedure	Section 7.3.1 or 7.3.2	B	B	
Crash Safety Sustained	Section 7.3.3	B	B	
Vibration	Section 8.5.2, Curve C	S	S	
Waterproofness	Section 10.2, 10.3.2	W	W	
Fluids Susceptibility, Spray Test	Section 11.4.1	F**	F**	
Fungus Resistance (by analysis)	Section 13	F	F	
Magnetic Effect	Section 15.3	B	B	
Electrical Power Input (ac)	Section 16.5	N/A	A(WF)	
Electrical Power Input (dc)	Section 16.6	Z	N/A	
Allowable Phase Unbalance (Three phase ac input only)	Section 16.7.2	N/A	A(WF)	

Test Description	DO-160 Test Section	Proposed Test Category (DC Eq.)	Proposed Test Category (AC Equip.)	Vendor Comments*
DC Current Content in Steady State	Section 16.7.3	N/A	A(WF)	
Inrush Current Requirements (ac & dc), Designation I	Section 16.7.5	I	I	
Current Modulation in Steady, State Operation (a), Designation L, Power > 35VA Equipt, or > 150VA total Equipt.	Section 16.7.6	N/A	L	
DC Current Ripple Tests (dc), Designation R, Power > 400W (all units), otherwise N/A	Section 16.7.7	R	N/A	
Power Factor P = PFC, Z = no correction	Section 16.7.8	N/A	P, Z	
Voltage Spike	Section 17.4	A	A	
AF Conducted Suscept., DC Input Power Leads	Section 18.3.1	Z	Z	
AF Conducted Suscept., AC Input Power Leads	Section 18.3.2	N/A	N/A	
Test Procedures, Magnetic Fields Induced Into the Equipment	Section 19.3.1	AW	AW	
Test Procedures, Electric Fields Induced Into the Equipment	Section 19.3.2	AW	AW	
Test Procedures, Magnetic Fields Induced Into Interconnecting Cables	Section 19.3.3	AW	AW	
Test Procedures, Electric Fields Induced Into Interconnecting Cables	Section 19.3.4	AW	AW	
Test Procedures, Spikes Induced Into Interconnecting Cables	Section 19.3.5	AW	AW	
Radio Freq. Susceptibility (Conducted)	Section 20.4	R	R	
Radio Freq. Susceptibility (Radiated)	Section 20.5	R	R	
EMI (Conducted RF Emission)	Section 21.4	M	M	
EMI (Radiated RF Emission)	Section 21.5	M	M	
Lightning Induced Transient Susceptibility	Section 22	B2H2XX***	B2H2XX***	

Test Description	DO-160 Test Section	Proposed Test Category (DC Eq.)	Proposed Test Category (AC Equip.)	Vendor Comments*
Electrostatic Discharge	**Section 25.5**	**A**	**A**	

*** Proposed Test Categories are part of the GTI qualification baseline, deviations will be considered on a case by case basis.**

**** Supplier to propose applicable fluids list**

***** B2 = Carbon Level 2 pin injection, H2 = Carbon level 2 cable injection, XX = No multiburst**

Appendix B - Vendor Data Requirements

Inventory Documents
List of Drawings Including: • Drawing numbers • Drawing descriptions or titles
List of Documents Including: • Document numbers • Document descriptions or titles
Drawings/Documents that Provide Component Definition for:
Center of gravity
Mass
Electrical connection(s)
Power requirement(s)
Cooling Requirement(s)
Dimensions
Mounting Provisions (interface)
All processes (finishes, coatings, heat treatments, bonding, etc.)
Mean Time Between/To Failure Data
Installation limitations
Installation instructions
Protection against pinch points, sharp edges, and/or trash traps
Identification of non-TSO items
Operation and Maintenance Documentation
Operation manuals
Repair procedures
Maintenance manuals
Illustrated parts catalogs
Electrical Information
Component Electrical Load Analyses
Conformity
TSO approval letter
PMA supplement
8130-3 Approval
Installation Approval
Supplemental Type Certificate (STC)
Structural Substantiation Data
Addressed within DO-160 Test reports
Flammability Documentation
Documentation showing compliance to 14 CFR 25.853(a), Amdt 25-116 and 25.869(a)(4) Amdt 25-113
Conformity records
FAA approvals
Additional Testing or Analysis

Any applicable testing and/or analysis, not described above, that was used to demonstrate compliance or qualify the component
Conformity records
FAA approvals

Appendix C- Control Matrix

	Zone	AeroSuite	Premium Lounge		Guest bedroom			Majlis		VIP Bedroom			Master Lavatory	Hallway	Business Class Seating		Economy Seating		Seat	Room
	Component Type	Passenger Control Panel	Room Control Panel	Passenger Control Panel	Room Control Panel	Bed Control Panel	Passenger Control Panel	Room Control Panel	Passenger Control Panel	Room Control Panel	Bed Control Panel	Passenger Control Panel	Room Control Panel	Room Control Panel	18.1" Seat Back Smart Monitor	Passenger Control Unit	11.6" Seatback Smart Monitor	Passenger Control Unit (1st Row only)	PED Tablets	PED Tablets
Control/Indication																				
Passenger Call		X		X		X	X		X		X	X	X		X	X	X	X	X	
Do Not Disturb		X																		
Room Call																			X	
Light Scenario		X	X		X	X		X		X	X		X	X					X	X
Individual Room Light		X	X		X	X		X		X	X		X	X						
Reading Light		X	X	X	X	X	X	X	X	X	X	X			X	X	X	X	X	
Shade Control		X	X	X	X	X	X	X	X	X	X	X	X	X					X	X
Monitor Power		X	X		X	X		X		X	X								X	X
Monitor controls **		X	X		X	X		X		X	X								X	X
Video Source Selection		X	X		X	X		X	X	X	X				X	X	X		X	X
VOD selection/Control		X	X		X	X		X		X	X				X	X	X		X	X
Audio Source Selection		X	X	X	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X
AOD selection/Control		X	X	X	X	X	X	X	X	X	X	X			X	X	X	X	X	X
Speaker/Headset Volume		X	X		X	X		X		X	X		X						X	X
Speaker Setting Control			X		X	X		X		X	X		X*						X	X
Headset Volume		X		X		X	X		X		X	X			X	X	X	X	X	
Moving Map w/ Control		X	X	X	X	X	X	X	X	X	X	X			X		X		X	X
Seat/Room Control			X		X	X		X		X	X									X

*Audio will be synced to the Master Bedroom **Monitor control shall include Subtitles, aspect ratio, contrast, Brightness

Appendix D- Audio/Video Matrix

Location	Monitors	Default On		A/VOD	eXTV	Moving Maps	HDMI 1 (Majlis)	HDMI 2 (VIP Bedroom)	HDMI 3 (Guest Bedroom)	HDMI 4 (Premium Lounge)	Broadcast Audio
Crew Terminal	(Qty 2)	Y		A/V*	A/V	V	A/V		A/V	A/V	
AeroSuite	32"	Y		A/V	A/V	V	A/V			A/V	A
Guest Bedroom	42"	Y		A/V	A/V	V	A/V		A/V	A/V	
Guest Bedroom Headphones				A	A		A		A	A	A
Premium Lounge	32"	Y		A/V	A/V	V	A/V			A/V	
Premium Lounge Headphones				A	A		A			A	A
Majlis	65"	Y		A/V	A/V	V	A/V			A/V	
Majlis Headphones				A	A		A			A	A
VIP Bedroom	65"	Y		A/V	A/V	V	A/V	A/V		A/V	
VIP Bedroom Headphones				A	A		A	A		A	A
VIP Bathroom				A	A		A	A		A	
Business Class	18"	Y		A/V	A/V	V				A/V	A
Economy Class	11"	Y		A/V	A/V	V					A

V* Video Controls /Preview only.

AeroSuite HDMI source is local only

Boarding Music = music played over PA speakers

CT default video is the CMI GUI.

Appendix E- Lighting Matrix

Lighting Scenarios are shown in "hue;saturation;value" (HSV) schema format. See following example below:

			Static Lighting Scenario			
Area	**Lights**	**Light State**	**Boarding**	**Dining**	**Movies**	**Sleeping**
AeroSuite	Accent Down (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**1%**	
	Toekick Lights	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**229;100;1**
		BRT; DIM 0%-100%; OFF	**100%**	**60%**	**1%**	**1%**
	Reading Lights	White	**No Change**	**No Change**	**No Change**	**No Change**

			Static Lighting Scenario			
Area	**Lights**	**Light State**	**Boarding**	**Dining**	**Movies**	**Sleeping**
Premium Lounge	Toekick Lights (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**229;100;1**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**1%**	**1%**
	Sconce lights	White	**BRT**	**DIM**	**OFF**	**OFF**
	Reading Lights	White	**No Change**	**No Change**	**No Change**	**No Change**

			Static Lighting Scenario			
Area	**Lights**	**Light State**	**Boarding**	**Dining**	**Movies**	**Sleeping**
Guest Bedroom	Dome Lights (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;0**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**0%**	
	Wash Lights (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**1%**	
	Ceiling Accent (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**1%**	
	Toekick Lights	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**229;100;1**
		BRT; DIM 0%-100%; OFF	**100%**	**60%**	**1%**	**1%**
	Sconce Lights	White	**100%**	**50%**	**OFF**	**OFF**
	Reading Lights	White	**No Change**	**No Change**	**No Change**	**No Change**

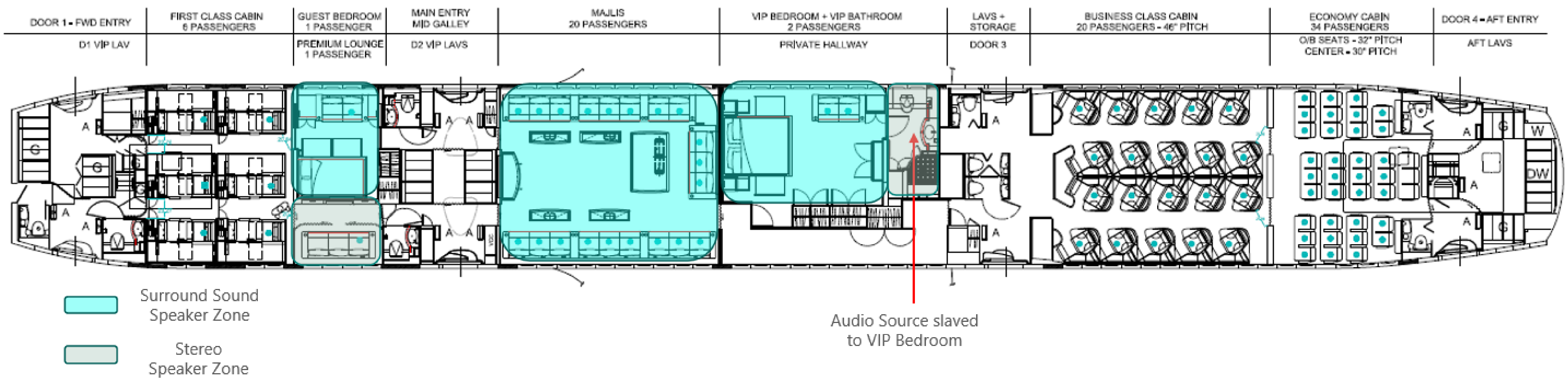
			Static Lighting Scenario			
Area	**Lights**	**Light State**	**Boarding**	**Dining**	**Movies**	**Sleeping**
Majlis	Dome Lights (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;0**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**0%**	
	Wash Lights (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**1%**	
	Ceiling Accent (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**1%**	
	Toekick Lights	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**229;100;1**
		BRT; DIM 0%-100%; OFF	**100%**	**60%**	**1%**	**1%**
	Accent Dome Lights	White	**100%**	**50%**	**OFF**	**OFF**
	Monitor Accent lights	White	**No Change**	**No Change**	**OFF**	**OFF**
	Reading Lights	White	**No Change**	**No Change**	**No Change**	**No Change**

			Static Lighting Scenario			
Area	**Lights**	**Light State**	**Boarding**	**Dining**	**Movies**	**Sleeping**
VIP Bedroom	Dome Lights (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;0**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**0%**	
	Wash Lights (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**1%**	
	Ceiling Accent (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**1%**	
	Toekick Lights	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**229;100;1**
		BRT; DIM 0%-100%; OFF	**100%**	**60%**	**1%**	**1%**
	Sconce Lights	White	**100%**	**50%**	**OFF**	**OFF**
	Reading Lights	White	**No Change**	**No Change**	**No Change**	**No Change**

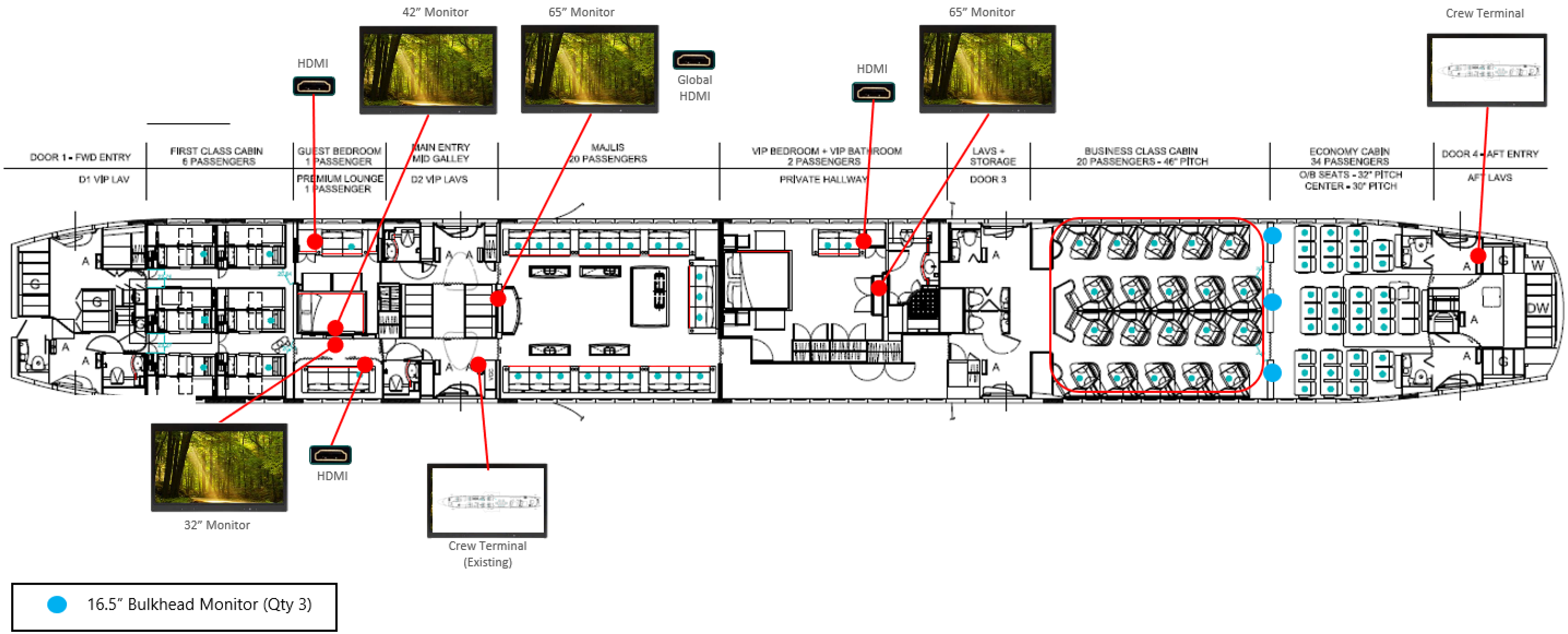
			Static Lighting Scenario			
Area	**Lights**	**Light State**	**Boarding**	**Dining**	**Movies**	**Sleeping**
VIP Bedroom	Dome Lights (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;0**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**0%**	
	Wash Lights (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**1%**	
	Ceiling Accent (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**1%**	
	Toekick Lights	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**229;100;1**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**1%**	**1%**
	Ceiling Dome Accent	White	**100%**	**50%**	**OFF**	**OFF**
	Mirror Lights	White	**No Change**	**No Change**	**No Change**	**OFF**
	Shower Lights	White	**No Change**	**No Change**	**No Change**	**No Change**

			Static Lighting Scenario			
Area	**Lights**	**Light State**	**Boarding**	**Dining**	**Movies**	**Sleeping**
Hallway	Dome Lights (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;0**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**0%**	
	Wash Lights (RGB)	RGB	**34;35;98**	**7;54;100**	**0;46;1**	**OFF**
		BRT; DIM 0%-100%; OFF	**98%**	**100%**	**1%**	

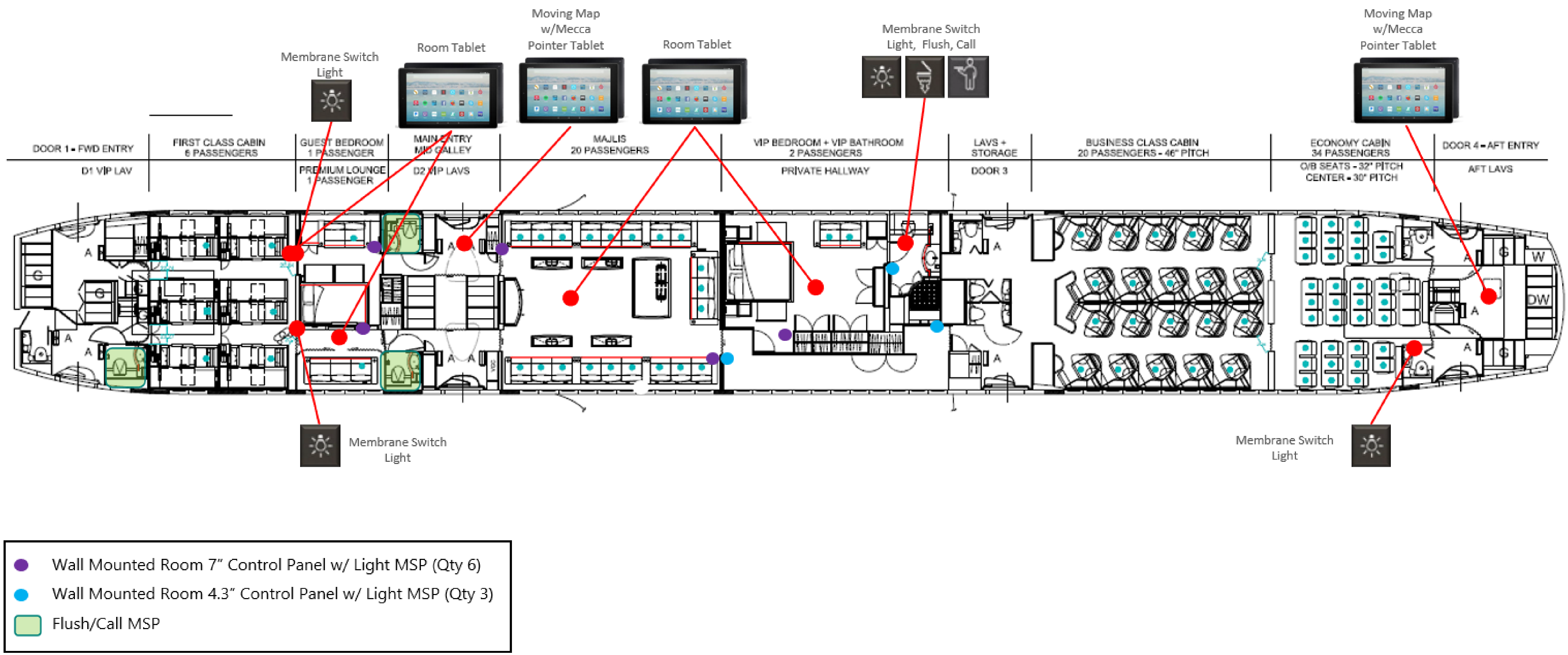
Appendix F- Component Locations



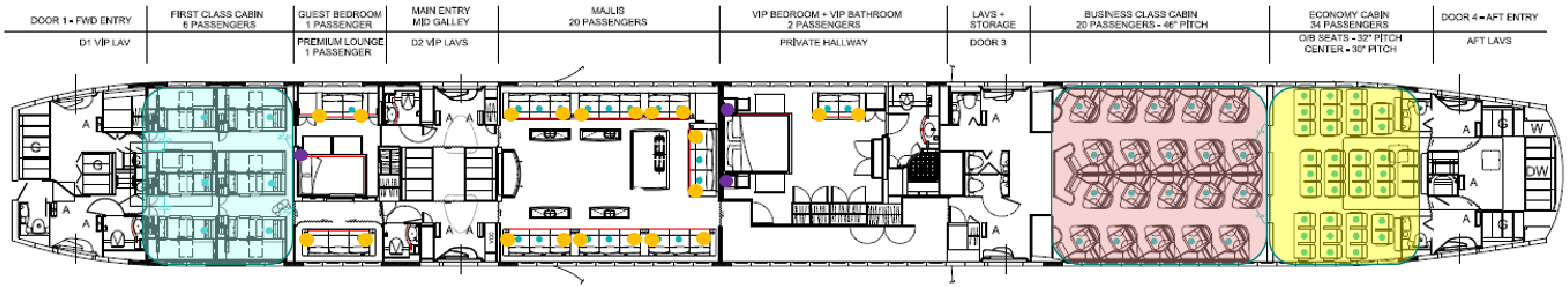
Entertainment Speakers



IFE/CMS Source and Display

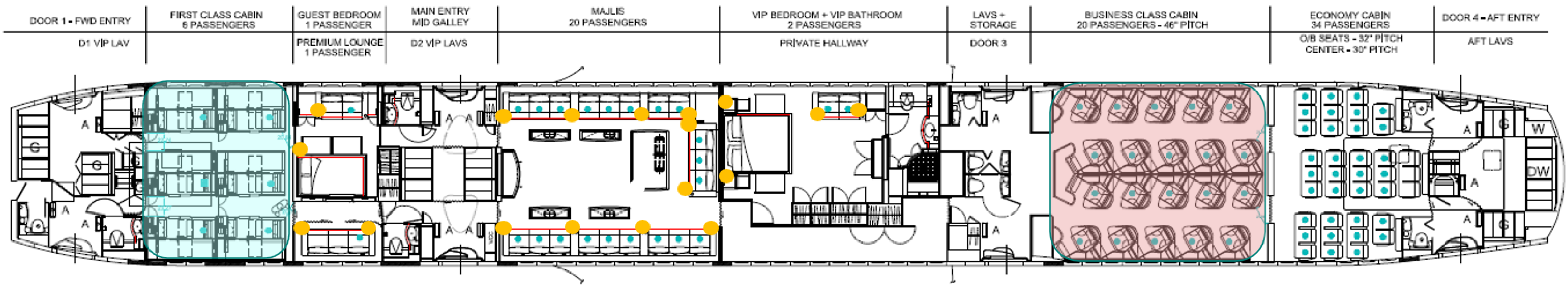


IFE/CMS Room Control



- 4.3" Passenger Control Unit W/Headphone Jack (Qty 18)
- 7" Passenger Control Unit w/Headphone Jack (Qty 3)
- Aero Suite contain 32" monitor, HDMI Port, 7" PCU, Headphone jack (Qty 6 total)
- 18.5" Smart monitor, Karma Handset W/Headphone Jack (20 Total) Existing Equipment
- 11.6" Monitor W/Headphone Jack & USB Port. (37 Total) Existing Equipment

IFE/CMS Seat Controls



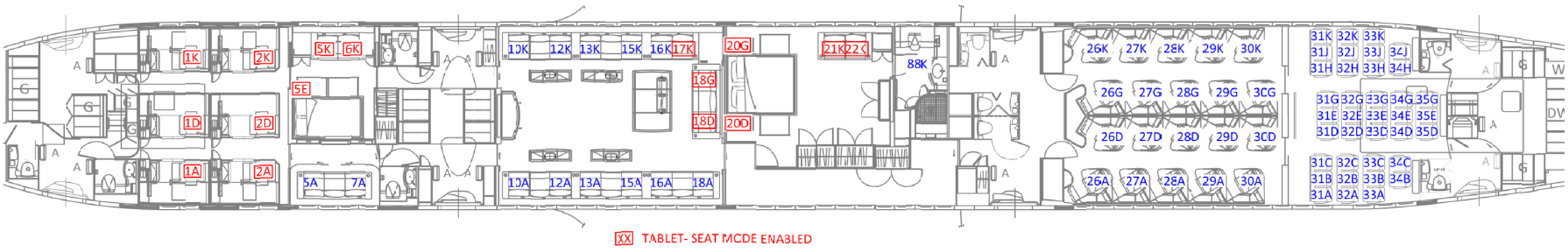
ISPS Outlet w/ USB Charging Port (Qty 18)

ISPS Outlet w/USB-C (6 Total)
USB-C Charging Port (6 Total)

ISPS Outlet and USB Charging Port (20 Total) Relocated Equipment

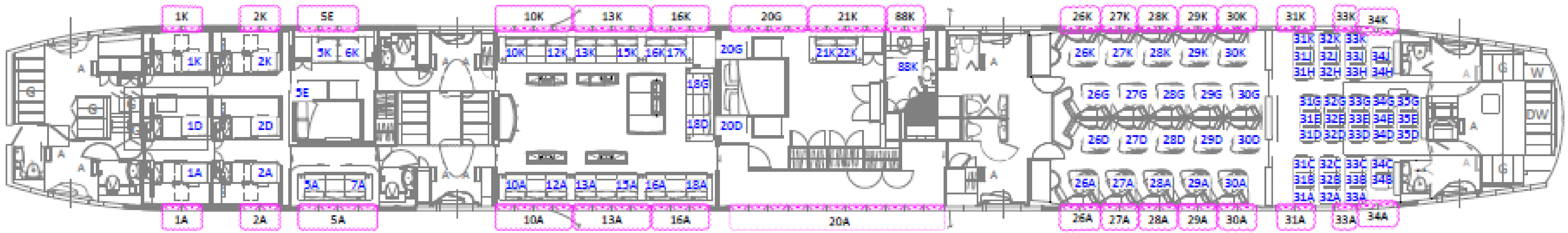
In-Seat Power Outlets

Appendix G- Seat LOPA Map



Seat LOPA Map

Appendix H Electrical Dimming Window Grouping



Electrical Dimming Window Grouping Map